

Am25LS192 • Am25LS193 Am54LS/74LS192 • Am54LS/74LS193

Decimal and Hexadecimal Up/Down Counters

DISTINCTIVE CHARACTERISTICS

- Separate up and down clocks
- Asynchronous parallel load
- Am25LS devices offer the following improvements over Am54/74LS
 - Higher speed
 - 50mV lower V_{OL} at $I_{OL} = 8\text{mA}$
 - Twice the fan-out over military range
 - 440 μA source current at HIGH output
- 100% product assurance screening to MIL-STD-883 requirements

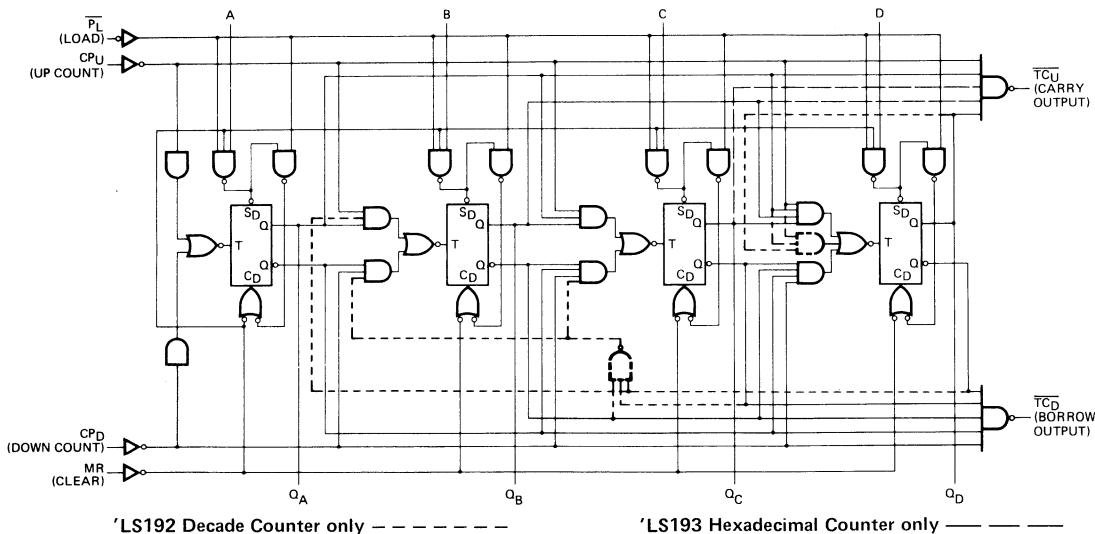
FUNCTIONAL DESCRIPTION

The 'LS192 and 'LS193 are four-bit up/down counters using advanced Low-Power Schottky processing. The 'LS192 counts in the BCD mode and the 'LS193 counts in the binary mode. These counters have separate count-up and count-down clock inputs (CP_U and CP_D , respectively). The Q_i outputs change state synchronously on the LOW-to-HIGH transition on either the up clock input or the down clock input. Only one clock input can be LOW at a time or erroneous counting will result.

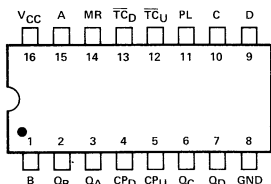
Each of the four flip-flops can be preset to a logic HIGH or a logic LOW by means of four parallel inputs (A, B, C, and D). When the parallel load input ($\overline{PL}$) goes LOW, all four flip-flops set to the state of the direct inputs (A, B, C, and D) independent of the clock inputs. An active HIGH master reset (MR) is provided which overrides both the clock and parallel load inputs forcing all Q_i outputs LOW.

Two terminal count outputs are gated with the clock inputs to provide clock signal to other counters. The TC_D output goes LOW when the counter is in state 0000 and the count down clock goes LOW. The TC_U goes LOW when the count up goes LOW and the counter is in state 1001 ('LS192) or state 1111 ('LS193). The TC_U and TC_D outputs can drive the count up and count down clocks on the next counter in a series. The Q_i outputs of such a connection scheme are not synchronous on cascaded counters in this series.

LOGIC DIAGRAM

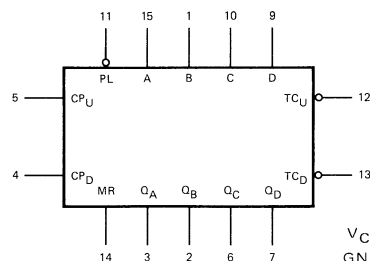


CONNECTION DIAGRAM Top View



Note: Pin 1 is marked for orientation.

LOGIC SYMBOL



V_{CC} = Pin 16
GND = Pin 8

Am25LS192 • Am25LS193

ELECTRICAL CHARACTERISTICS OVER OPERATING TEMPERATURE RANGE (Unless Otherwise Noted)

Am25LS192XC/Am25LS193XC	$T_A = 0^{\circ}\text{C to } +70^{\circ}\text{C}$	$V_{CC} = 5.0\text{V} \pm 5\% \text{ (COM'L)}$	MIN. = 4.75V	MAX. = 5.25V
Am25LS192XM/Am25LS193XM	$T_A = -55^{\circ}\text{C to } +125^{\circ}\text{C}$	$V_{CC} = 5.0\text{V} \pm 10\% \text{ (MIL)}$	MIN. = 4.50V	MAX. = 5.50V

Parameters	Description	Test Conditions (Note 1)		Min.	Typ. (Note 2)	Max.	Units
V _{OH}	Output HIGH Voltage	V _{CC} = MIN., I _{OH} = −440μA V _{IN} = V _{IH} or V _{IL}	MIL	2.5	3.4		Volts
			COM'L	2.7	3.4		
V _{OL}	Output LOW Voltage	V _{CC} = MIN., V _{IN} = V _{IH} or V _{IL}	I _{OL} = 4.0mA		0.25	0.40	Volts
			I _{OL} = 8.0mA		0.35	0.45	
V _{IH}	Input HIGH Level	Guaranteed input logical HIGH voltage for all inputs		2.0			Volts
V _{IL}	Input LOW Level	Guaranteed input logical LOW voltage for all inputs		MIL		0.7	Volts
				COM'L		0.8	
V _I	Input Clamp Voltage	V _{CC} = MIN., I _{IN} = −18mA				−1.5	Volts
I _{IL}	Input LOW Current	V _{CC} = MAX., V _{IN} = 0.4 V				−0.4	mA
I _{IH}	Input HIGH Current	V _{CC} = MAX., V _{IN} = 2.7 V				20	μA
I _I	Input HIGH Current	V _{CC} = MAX., V _{IN} = 7.0 V				0.1	mA
I _{SC}	Output Short Circuit Current (Note 3)	V _{CC} = MAX.		−15		−85	mA
I _{CC}	Power Supply Current	V _{CC} = MAX. (Note 4)			19	34	mA

Notes: 1. For conditions shown as MIN. or MAX., use the appropriate value specified under Electrical characteristics for the applicable device type.

2. Typical limits are at $V_{CC} = 5.0\text{V}$, 25°C ambient and maximum loading.

3. Not more than one output should be shorted at a time. Duration of the short circuit test should not exceed one second.

4. I_{CC} is measured with all outputs open; clear and load inputs grounded; and all other inputs at 4.5V

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Am25LS • Am54LS/74LS

MAXIMUM RATINGS (Above which the useful life may be impaired)

Storage Temperature	$-65^{\circ}\text{C to } +150^{\circ}\text{C}$
Temperature (Ambient) Under Bias	$-55^{\circ}\text{C to } +125^{\circ}\text{C}$
Supply Voltage to Ground Potential Continuous	$-0.5\text{V to } +7.0\text{V}$
DC Voltage Applied to Outputs for High Output State	$-0.5\text{V to } +V_{CC} \text{ max.}$
DC Input Voltage	$-0.5\text{V to } +7.0\text{V}$
DC Output Current, Into Outputs	30mA
DC Input Current	$-30\text{mA to } +5.0\text{mA}$

Am25LS/54LS/74LS192/193

Am54LS/74LS192 • Am54LS/74LS193

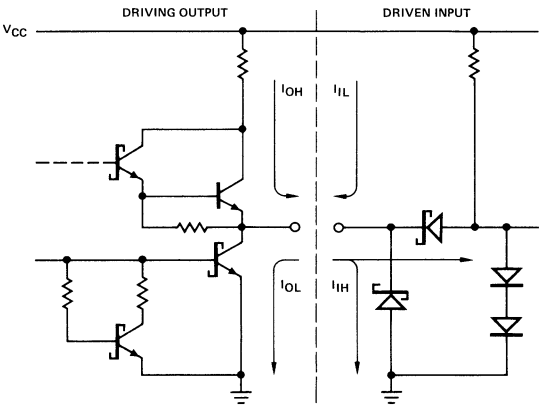
ELECTRICAL CHARACTERISTICS OVER OPERATING TEMPERATURE RANGE (Unless Otherwise Noted)

Am74LS192X/74LS193X	$T_A = 0^{\circ}\text{C to } +70^{\circ}\text{C}$	$V_{CC} = 5.0\text{V} \pm 5\% \text{ (COM'L)}$	MIN. = 4.75V	MAX. = 5.25V
Am54LS192X/54LS193X	$T_A = -55^{\circ}\text{C to } +125^{\circ}\text{C}$	$V_{CC} = 5.0\text{V} \pm 10\% \text{ (MIL)}$	MIN. = 4.50V	MAX. = 5.50V

Parameters	Description	Test Conditions (Note 1)	Min.	Typ. (Note 2)	Max.	Units
V_{OH}	Output HIGH Voltage	$V_{CC} = \text{MIN.}, I_{OH} = -400\mu\text{A}$ $V_{IN} = V_{IH} \text{ or } V_{IL}$	MIL 2.5 COM'L 2.7	3.4 3.4		Volts
V_{OL}	Output LOW Voltage	$V_{CC} = \text{MIN.},$ $V_{IN} = V_{IH} \text{ or } V_{IL}$	All, $I_{OL} = 4.0\text{mA}$ 74LS only, $I_{OL} = 8.0\text{mA}$	0.25 0.35	0.40 0.50	Volts
V_{IH}	Input HIGH Level	Guaranteed input logical HIGH voltage for all inputs	2.0			Volts
V_{IL}	Input LOW Level	Guaranteed input logical LOW voltage for all inputs	MIL COM'L		0.7 0.8	Volts
V_I	Input Clamp Voltage	$V_{CC} = \text{MIN.}, I_{IN} = -18\text{mA}$			-1.5	Volts
I_{IL}	Input LOW Current	$V_{CC} = \text{MAX.}, V_{IN} = 0.4\text{V}$			-0.4	mA
I_{IH}	Input HIGH Current	$V_{CC} = \text{MAX.}, V_{IN} = 2.7\text{V}$			20	μA
I_I	Input HIGH Current	$V_{CC} = \text{MAX.}, V_{IN} = 7.0\text{V}$			0.1	mA
I_{SC}	Output Short Circuit Current (Note 3)	$V_{CC} = \text{MAX.}$	-15		-100	mA
I_{CC}	Power Supply Current	$V_{CC} = \text{MAX.}$ (Note 4)		19	34	mA

- Notes: 1. For conditions shown as MIN. or MAX., use the appropriate value specified under Electrical Characteristics for the applicable device type.
2. Typical limits are at $V_{CC} = 5.0\text{V}$, 25°C ambient and maximum loading.
3. Not more than one output should be shorted at a time. Duration of the short circuit test should not exceed one second.
4. I_{CC} is measured with all outputs open; clear and load inputs grounded; and all other inputs at 4.5V

Am25LS • 54LS/74LS
LOW-POWER SCHOTTKY INPUT/OUTPUT
CURRENT INTERFACE CONDITIONS



Note: Actual current flow direction shown.

SWITCHING CHARACTERISTICS(T_A = 25°C, V_{CC} = 5.0V)

Parameters		Description		Min.	Typ.	Max.	Min.	Typ.	Max.	Units	Test Conditions
t _{PLH}	CP _U or CP _D to Q _n			24	34		25	38		ns	C _L = 15pF R _L = 2.0kΩ
t _{PHL}				28	39		31	47			
t _{PLH}	CP _U to TC _U			10	15		17	26		ns	
t _{PHL}				10	15		21	33			
t _{PLH}	CP _D to TC _D			10	15		16	24		ns	
t _{PHL}				11	17		21	33			
t _{PLH}	A-D to Q _n Output			13	18			—		ns	
t _{PHL}				27	38			—			
t _{PLH}	A-D to TC _U Output			35	49			—		ns	
t _{PHL}				19	27			—			
t _{PLH}	A-D to TC _D Output			26	36			—		ns	
t _{PHL}				28	39			—			
t _{PHL}	MR Input to Q _n Output			20	29		22	35		ns	
t _{PLH}	MR Input to TC _U Output			25	35			—		ns	
t _{PHL}	MR Input to TC _D Output			16	22			—		ns	
t _{PLH}	P _L Input to Q _n Output			20	29		27	40		ns	
t _{PHL}				25	36		29	40			
t _{PLH}	P _L Input to TC _U Output			31	45			—		ns	
t _{PHL}				30	42			—			
t _{PLH}	P _L Input to TC _D Output			30	42			—		ns	
t _{PHL}				24	34			—			
t _s	Data Set-up Time A-D Input to P _L Input	Load 1	5.0				—			ns	
		Load 0	15			20				ns	
t _s	Set-up Time, P _L Input to CP _U or CP _D		9.0			—				ns	
t _s	Set-up Time, Clear Recovery (In-Active) to CP _U or CP _D		5.0			—				ns	
t _h	Data		0			0				ns	
t _{pw(0)}	Pulse Width	CP _U	11			20				ns	
		CP _D	11			20					
		P _L	9.0			20					
t _{pw(1)}	Pulse Width	MR	15			20				ns	
f _{max}	Maximum Clock Frequency, Count Up or Down (Note 1)		35	45		25	32			MHz	

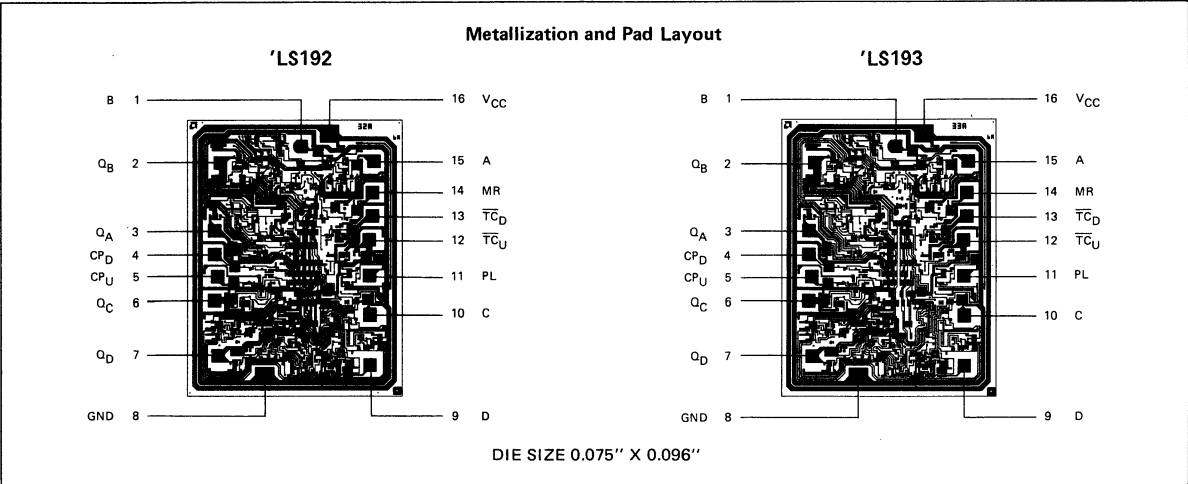
Note 1. Per industry convention, f_{max} is the worst case value of the maximum device operating frequency with no constraints on t_r, t_f, pulse width or duty cycle.

Am25LS/54LS/74LS192/193

Am25LS192, Am25LS193 ONLY
SWITCHING CHARACTERISTICS
OVER OPERATING RANGE*

Am25LS192, Am25LS193 ONLY SWITCHING CHARACTERISTICS OVER OPERATING RANGE*			Am25LS COM'L		Am25LS MIL		Units	Test Conditions
Parameters			Description		T _A = 0°C to +70°C V _{CC} = 5.0V ± 5%			
			Min.	Max.	Min.	Max.		
t _{PLH}	CP _U or CP _D to Q _n			44		52	ns	C _L = 50pF R _L = 2.0kΩ
t _{PHL}				54		67		
t _{PLH}	CP _U to TC _U			22		26	ns	
t _{PHL}				23		27		
t _{PLH}	CP _D to TC _D			22		26	ns	
t _{PHL}				23		27		
t _{PLH}	A-D to Q _n Output			26		31	ns	
t _{PHL}				52		63		
t _{PLH}	A-D to TC _U Output			66		80	ns	
t _{PHL}				38		46		
t _{PLH}	A-D to TC _D Output			50		60	ns	
t _{PHL}				54		66		
t _{PHL}	MR Input to Q _n Output			41		50	ns	
t _{PLH}	MR Input to TC _U Output			49		60	ns	
t _{PHL}	MR Input to TC _D Output			32		38	ns	
t _{PLH}	P _L Input to Q _n Output			41		50	ns	
t _{PHL}				53		67		
t _{PLH}	P _L Input to TC _U Output			63		79	ns	
t _{PHL}				60		75		
t _{PLH}	P _L Input to TC _D Output			60		75	ns	
t _{PHL}				48		60		
t _s	Data Set-up Time A-D Input to P _L Input	Load 1	5.0		5.0	ns		
		Load 0	20		23			
t _s	Set-up Time, P _L Input to CP _U or CP _D		13		18	ns		
t _s	Set-up Time, Clear Recovery (In-Active) to CP _U or CP _D		7.0		9.0	ns		
t _h	Data		0		0	ns		
t _{pw(0)}	Pulse Width	CP _U	15		17	ns		
		CP _D	15		17			
		P _L	13		17			
t _{pw(1)}	Pulse Width	MR	18		21			
f _{max}	Maximum Clock Frequency, Count Up or Down (Note 1)		25		20	MHz		

* AC performance over the operating temperature range is guaranteed by testing defined in Group A, Subgroup 9.



FUNCTION TABLE

INPUTS								OUTPUTS						Conditions
Clock				Data										
Up	Down	Clear	Load	A	B	C	D	Q _A	Q _B	Q _C	Q _D	Borrow	Carry	
X X	L H	H H	X X	X X	X X	X X	X X	L L	L L	L L	L L	L H	H H	Clear
X	X	L	L	D ₀	D ₁	D ₂	D ₃	D ₀	D ₁	D ₂	D ₃	X	X	
H	↑	L	H	X	X	X	X	Count Down				H	H	Except at borrow
H H	L H	L L	H H	X X	X X	X X	X X	L L	L L	L L	L L	L H	H H	Borrow
↑	H	L	H	X	X	X	X	Count up				H	H	
L H	H H	L L	H H	X X	X X	X X	X X	H H	H H	H H	H H	H H	L H	Carry (193 only)
L H	H H	L L	H H	X X	X X	X X	X X	H H	L L	L L	H H	H H	L H	

H = HIGH X = Don't care D = A LOW or a HIGH and the respective output will assume the same state.
L = LOW ↑ = LOW-to-HIGH transition

DEFINITION OF FUNCTIONAL TERMS

- MR** Clear. The clear input to the counter overrides all other inputs. When the clear input is HIGH, the Q outputs are set LOW independent of the other inputs.

PL Load. The load input performs asynchronous parallel load of the data on the A, B, C, and D inputs. When the load input is LOW, the Q_i outputs will follow the parallel inputs regardless of the clock inputs.

A, B, C, D The four parallel inputs to the counter flip-flops.

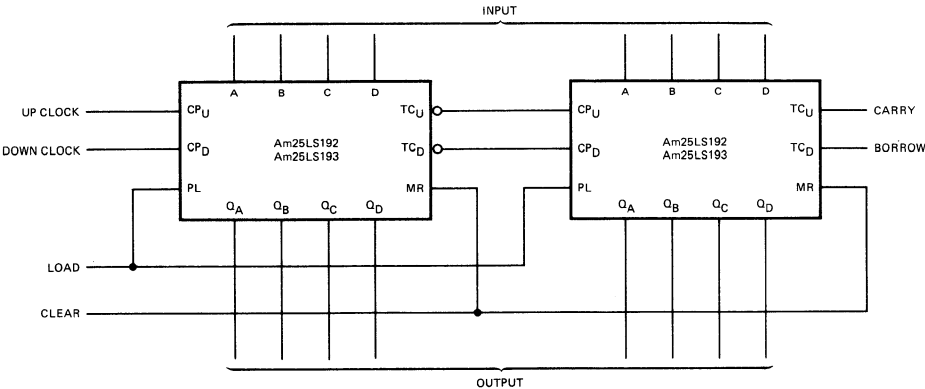
CP_U Count up. A clock input causing the counter to change state in an increasing binary number direction. Counting occurs on the LOW-to-HIGH transition of the clock.
- CP_D** Count down. A clock input causing the counter to change state in a decreasing binary number direction. The state change occurs on the LOW-to-HIGH transition.

Q_A, Q_B, Q_C, Q_D The four outputs of the counter representing the LSB to MSB, respectively.

TC_U Carry output. A clock output that indicates the maximum upper binary number has been reached. For the 'LS192, TC_U indicates that the "9" state has been reached and the up clock is LOW. For the 'LS193, TC_U indicates that the "15" state has been reached and the up clock is LOW.

TC_D Borrow output. A clock output indicating that the "0" state has been reached and the down clock is LOW.

APPLICATION



8-Bit Up/Down Counter with Parallel Load